Math F113X: Quiz 5 17 October, 2025

Name: _____ score:_____ / 10

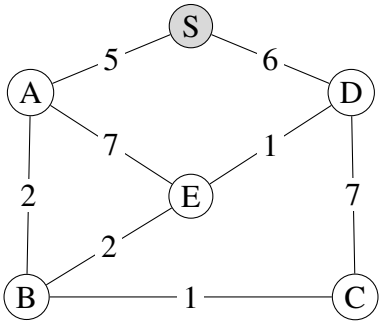
There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a non-programmable calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

1. (4 points) Recall Dijkstra's algorithm says the following:

   Dijkstra's Algorithm

   input: a graph with distances (weights) on the edges and a starting vertex, say s
   output: the shortest distance between s and every vertex in the graph
   rough strategy: All vertices get **tentative** distances to vertex s. One-by-one, vertices are explored and tentative distances are updated until minimum distances are obtained. Break ties alphabetically.

   Use **Dijkstra's algorithm** to determine the distances between vertex S and each other vertex. Clearly show the steps of the algorithm in the space provided.



Explored?	vertices	tentative distances
	S	
	A	
	B	
	C	
	D	
	E	

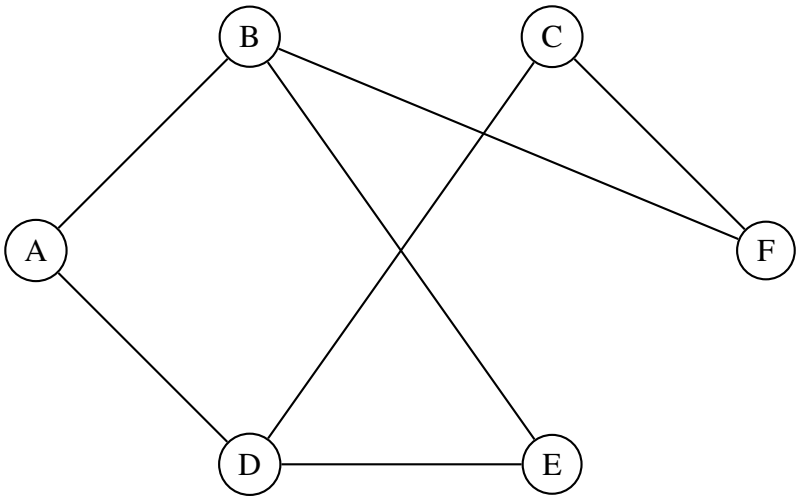
List the vertices in the order you explored them.

_____

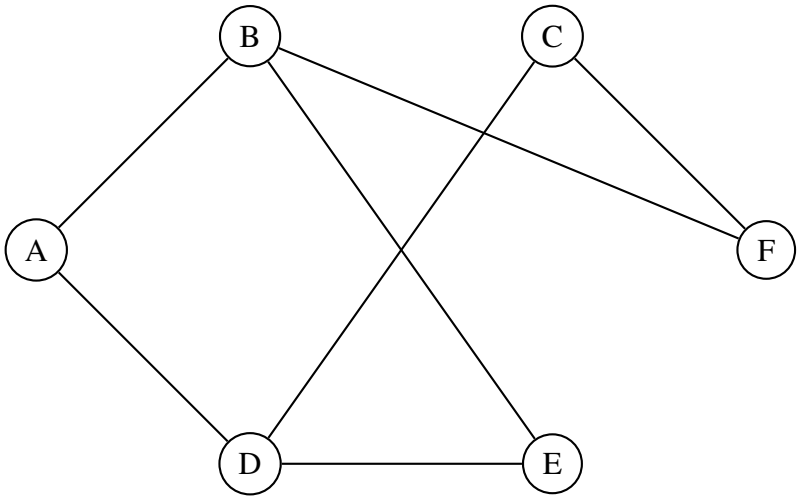
List the **final** distances to S.

vertex	S	A	B	C	D	E
distance from S						

2. (6 points) Consider the following graph.



(a) Explain why this graph does not have an **Euler circuit**.

(b) Add additional edges to **Eulerize** the graph above.

(c) Add the additional edges to the copy of the graph below and draw an **Euler circuit** on the graph. Offset it slightly from the graph so the order it is traveled in is clear.